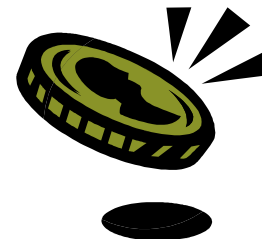


Bayesian Belief Networks

Probability Primer: Random Variables

- A **random variable** is the basic element of probability
- Refers to an event and there is some degree of uncertainty as to the outcome of the event
- For example, the random variable A could be the event of getting a head on a coin flip



Boolean Random Variables

- We will start with the simplest type of random variables – Boolean ones
- Take the values *true* or *false*
- Think of the event as occurring or not occurring
- Examples (Let A be a Boolean random variable):
 - A = Getting a head on a coin flip
 - A = It will rain today

The Joint Probability Distribution

- Joint probabilities can be between any number of variables
eg. $P(A = \text{true}, B = \text{true}, C = \text{true})$
- For each combination of variables, we need to say how probable that combination is
- The probabilities of these combinations need to sum to 1

A	B	C	P(A,B,C)
false	false	false	0.1
false	false	true	0.2
false	true	false	0.05
false	true	true	0.05
true	false	false	0.3
true	false	true	0.1
true	true	false	0.05
true	true	true	0.15

Sums to 1

The Joint Probability Distribution

- Once you have the joint probability distribution, you can calculate any probability involving A , B , and C
- Note: May need to use marginalization and Bayes rule, (both of which are not discussed in these slides)

A	B	C	P(A,B,C)
false	false	false	0.1
false	false	true	0.2
false	true	false	0.05
false	true	true	0.05
true	false	false	0.3
true	false	true	0.1
true	true	false	0.05
true	true	true	0.15

Examples of things you can compute:

- $P(A=true) = \text{sum of } P(A,B,C) \text{ in rows with } A=true$
- $P(A=true, B = true \mid C=true) =$
 $P(A = true, B = true, C = true) / P(C = true)$

The Problem with the Joint Distribution

- Lots of entries in the table to fill up!
- For k Boolean random variables, you need a table of size 2^k
- How do we use fewer numbers? Need the concept of independence

A	B	C	P(A,B,C)
false	false	false	0.1
false	false	true	0.2
false	true	false	0.05
false	true	true	0.05
true	false	false	0.3
true	false	true	0.1
true	true	false	0.05
true	true	true	0.15

Independence

Variables A and B are independent if any of the following hold:

- $P(A, B) = P(A) P(B)$
- $P(A \mid B) = P(A)$
- $P(B \mid A) = P(B)$

This says that knowing the outcome of A does not tell me anything new about the outcome of B .

Independence

How is independence useful?

- Suppose you have n coin flips and you want to calculate the joint distribution $P(C_1, \dots, C_n)$
- If the coin flips are not independent, you need 2^n values in the table
- If the coin flips are independent, then

$$P(C_1, \dots, C_n) = \prod_{i=1}^n P(C_i)$$

Each $P(C_i)$ table has 2 entries and there are n of them for a total of $2n$ values

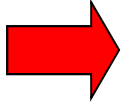
Conditional Independence

Variables A and B are conditionally independent given C if any of the following hold:

- $P(A, B \mid C) = P(A \mid C) P(B \mid C)$
- $P(A \mid B, C) = P(A \mid C)$
- $P(B \mid A, C) = P(B \mid C)$

Knowing C tells me everything about B . I don't gain anything by knowing A (either because A doesn't influence B or because knowing C provides all the information knowing A would give)

Outline

1. Introduction
2. Probability Primer
-  3. Bayesian networks

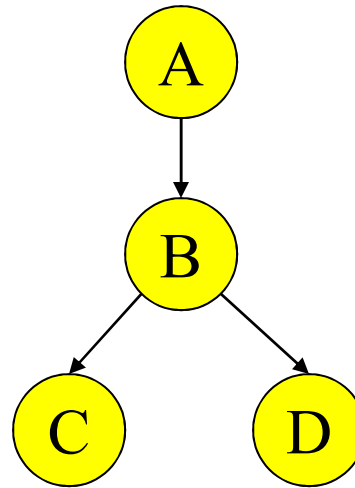
Bayesian networks or Bayesian Belief Network (BBN)

- A simple, graphical notation for conditional independence assertions and hence for compact specification of full joint distributions
- Syntax:
 - a set of nodes, one per variable
 - a directed, acyclic graph (link $\approx$ "directly influences")
 - a conditional distribution for each node given its parents:
$$\mathbf{P}(X_i \mid \text{Parents}(X_i))$$
- In the simplest case, conditional distribution represented as a **conditional probability table** (CPT) giving the distribution over X_i for each combination of parent values
- A node is independent of its nondescendants given its parents.

A Bayesian Network

A Bayesian network is made up of:

1. A Directed Acyclic Graph



2. A set of tables for each node in the graph

A	P(A)
false	0.6
true	0.4

A	B	P(B A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

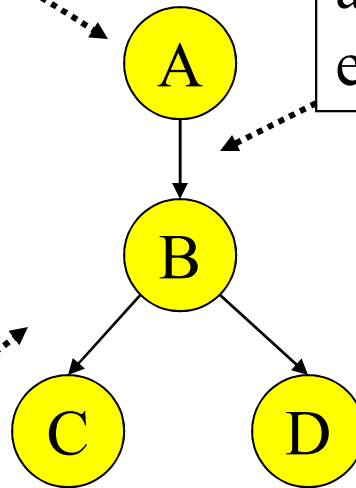
B	D	P(D B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

B	C	P(C B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

A Directed Acyclic Graph

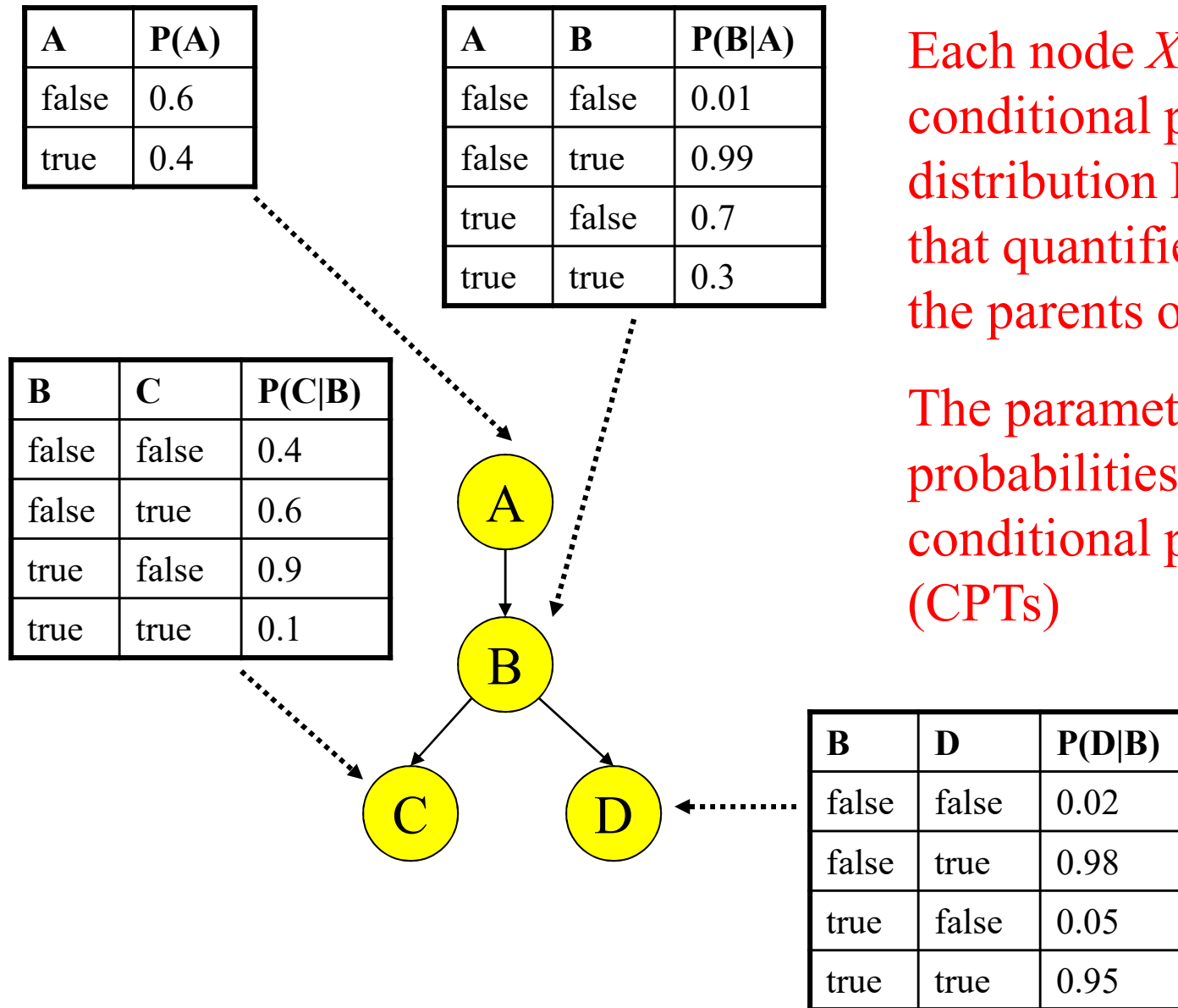
Each node in the graph is a random variable

A node X is a parent of another node Y if there is an arrow from node X to node Y
eg. A is a parent of B



Informally, an arrow from node X to node Y means X has a direct influence on Y

A Set of Tables for Each Node



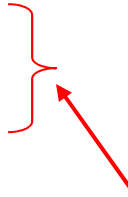
Each node X_i has a conditional probability distribution $P(X_i \mid \text{Parents}(X_i))$ that quantifies the effect of the parents on the node

The parameters are the probabilities in these conditional probability tables (CPTs)

A Set of Tables for Each Node

Conditional Probability
Distribution for C given B

B	C	P(C B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1



For a given combination of values of the parents (B in this example), the entries for $P(C=\text{true} \mid B)$ and $P(C=\text{false} \mid B)$ must add up to 1
eg. $P(C=\text{true} \mid B=\text{false}) + P(C=\text{false} \mid B=\text{false}) = 1$

If you have a Boolean variable with k Boolean parents, this table has 2^{k+1} probabilities (but only 2^k need to be stored)

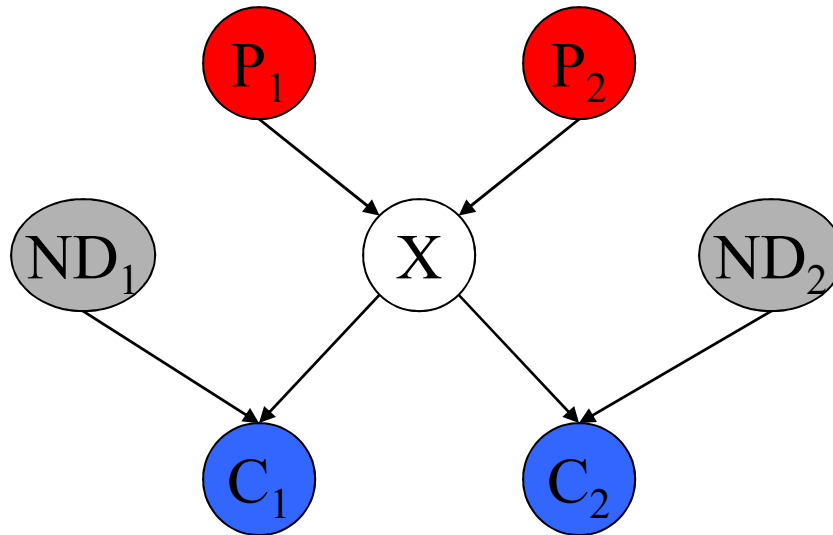
Bayesian Networks

Two important properties:

1. Encodes the conditional independence relationships between the variables in the graph structure
2. Is a compact representation of the joint probability distribution over the variables

Conditional Independence

The Markov condition: given its parents (P_1, P_2), a node (X) is conditionally independent of its non-descendants (ND_1, ND_2)



The Joint Probability Distribution

Due to the Markov condition, we can compute the joint probability distribution over all the variables $X_1, \dots, X_n$ in the Bayesian net using the formula:

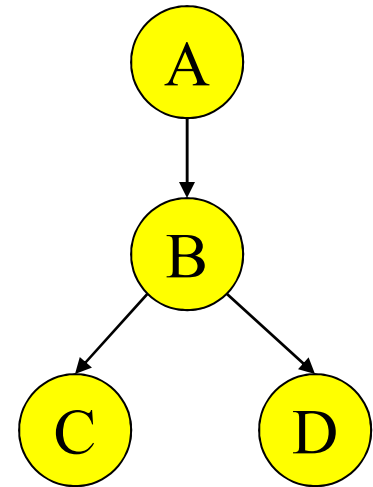
$$P(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n P(X_i = x_i \mid \text{Parents}(X_i))$$

Where $\text{Parents}(X_i)$ means the values of the Parents of the node X_i with respect to the graph

Using a Bayesian Network Example

Using the network in the example, suppose you want to calculate:

$$\begin{aligned} &P(A = \text{true}, B = \text{true}, C = \text{true}, D = \text{true}) \\ &= P(A = \text{true}) * P(B = \text{true} \mid A = \text{true}) * \\ &\quad P(C = \text{true} \mid B = \text{true}) P(D = \text{true} \mid B = \text{true}) \\ &= (0.4)*(0.3)*(0.1)*(0.95) \end{aligned}$$



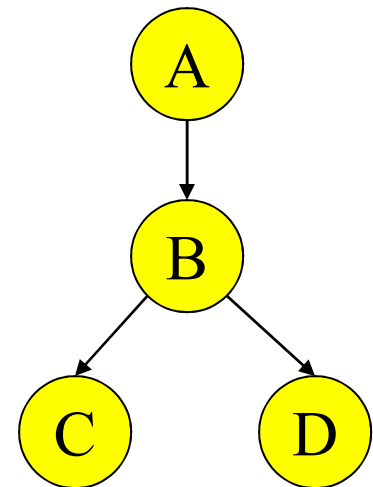
Using a Bayesian Network Example

Using the network in the example, suppose you want to calculate:

$$\begin{aligned} &P(A = \text{true}, B = \text{true}, C = \text{true}, D = \text{true}) \\ &= P(A = \text{true}) * P(B = \text{true} \mid A = \text{true}) * \\ &\quad P(C = \text{true} \mid B = \text{true}) P(D = \text{true} \mid B = \text{true}) \\ &= (0.4) * (0.3) * (0.1) * (0.95) \end{aligned}$$

This is from the
graph structure

These numbers are from the
conditional probability tables



Joint Probability Factorization

- Joint Probability Factorization is a key concept in probabilistic modeling, especially within the context of Bayesian Networks.
- It refers to the process of expressing the **joint probability distribution of a set of variables as a product of conditional probabilities based on the structure of the network.**

Mathematical Representation

- For a set of variables $X_1, X_2, \dots, X_n$, the joint probability $P(X_1, X_2, \dots, X_n)$ can be factorized as follows:
 - $P(X_1, X_2, \dots, X_n) = P(X_1) \cdot P(X_2|X_1) \cdot P(X_3|X_1, X_2) \cdots P(X_n|X_1, X_2, \dots, X_{n-1})$.
- This representation indicates that each variable can be expressed in terms of its parents in the network, thereby allowing the joint distribution to be broken down into simpler components.

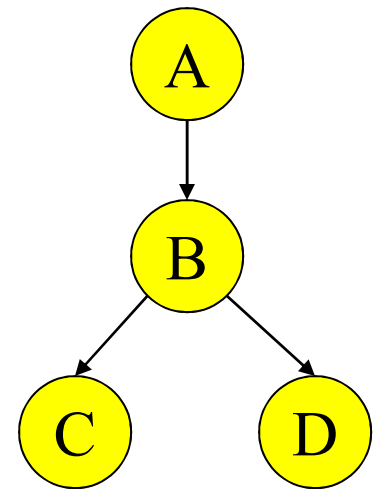
Joint Probability Factorization

Our example graph carries additional independence information, which simplifies the joint distribution:

$$\begin{aligned} P(A, B, C, D) &= P(A)P(B \mid A)P(C \mid A, B)P(D \mid A, B, C) \\ &= P(A)P(B \mid A)P(C \mid B)P(D \mid B) \end{aligned}$$

This is why, we only need the tables for $P(A)$, $P(B|A)$, $P(C|B)$, and $P(D|B)$ and why we computed

$$\begin{aligned} &P(A = \text{true}, B = \text{true}, C = \text{true}, D = \text{true}) \\ &= P(A = \text{true}) * P(B = \text{true} \mid A = \text{true}) * \\ &\quad P(C = \text{true} \mid B = \text{true}) P(D = \text{true} \mid B = \text{true}) \\ &= (0.4)*(0.3)*(0.1)*(0.95) \end{aligned}$$



Inference

- Using a Bayesian network to compute probabilities is called inference
- In general, inference involves queries of the form:

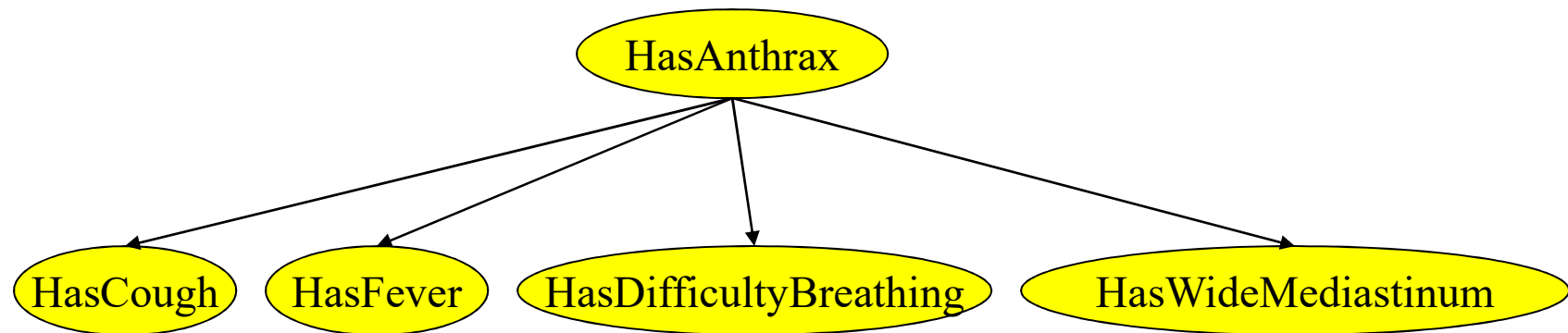
$$P(X \mid E)$$



E = The evidence variable(s)

X = The query variable(s)

Inference



- An example of a query would be:
 $P(\text{HasAnthrax} = \text{true} \mid \text{HasFever} = \text{true}, \text{HasCough} = \text{true})$
- Note: Even though *HasDifficultyBreathing* and *HasWideMediastinum* are in the Bayesian network, they are not given values in the query (ie. they do not appear either as query variables or evidence variables)
- They are treated as unobserved variables and summed out.

Inference Example

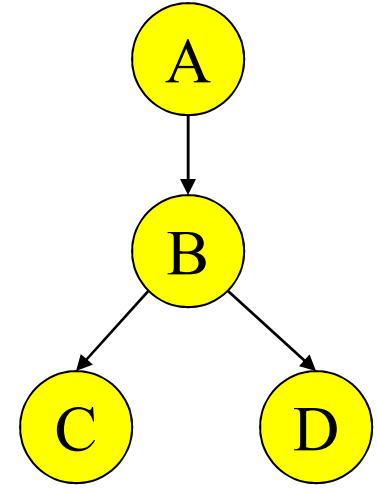
Supposed we know that $A=\text{true}$.

What is more probable $C=\text{true}$ or $D=\text{true}$?

For this we need to compute

$P(C=t \mid A=t)$ and $P(D=t \mid A=t)$.

Let us compute the first one.



$$P(C = t \mid A = t) = \frac{P(A = t, C = t)}{P(A = t)} = \frac{\sum_{b,d} P(A = t, B = b, C = t, D = d)}{P(A = t)}$$

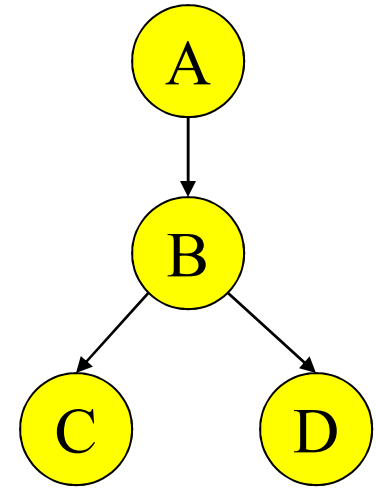
A	P(A)
false	0.6
true	0.4

A	B	P(B A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

B	D	P(D B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

B	C	P(C B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

What is $P(A=\text{true})$?



$$\begin{aligned}
 P(A=t) &= \sum_{b,c,d} P(A=t, B=b, C=c, D=d) \\
 &= \sum_{b,c,d} P(A=t)P(B=b \mid A=t)P(C=c \mid B=b)P(D=d \mid B=b) \\
 &= P(A=t) \sum_{b,c,d} P(B=b \mid A=t)P(C=c \mid B=b)P(D=d \mid B=b) \\
 &= P(A=t) \sum_b P(B=b \mid A=t) \sum_{c,d} P(C=c \mid B=b)P(D=d \mid B=b) \\
 &= P(A=t) \sum_b P(B=b \mid A=t) \sum_c P(C=c \mid B=b) \sum_d P(D=d \mid B=b) \\
 &= P(A=t) \sum_b P(B=b \mid A=t) \sum_c P(C=c \mid B=b) * 1 \\
 &= 0.4(P(B=t \mid A=t) \sum_c P(C=c \mid B=t) + P(B=f \mid A=t) \sum_c P(C=c \mid B=f)) = \dots
 \end{aligned}$$

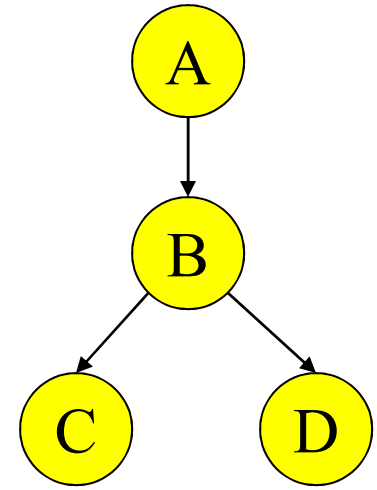
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false	true	0.99
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true	true	0.3

B	D	P(D B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

B	C	P(C B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

What is $P(C=\text{true}, A=\text{true})$?



$$\begin{aligned}
 P(A=t, C=t) &= \sum_{b,d} P(A=t, B=b, C=t, D=d) \\
 &= \sum_{b,d} P(A=t)P(B=b \mid A=t)P(C=t \mid B=b)P(D=d \mid B=b) \\
 &= P(A=t) \sum_b P(B=b \mid A=t)P(C=t \mid B=b) \sum_d P(D=d \mid B=b) \\
 &= 0.4(P(B=t \mid A=t)P(C=t \mid B=t) \sum_d P(D=d \mid B=t) \\
 &\quad + P(B=f \mid A=t)P(C=t \mid B=f) \sum_d P(D=d \mid B=f)) \\
 &= 0.4(0.3 * 0.1 * 1 + 0.7 * 0.6 * 1) = 0.4(0.03 + 0.42) = 0.4 * 0.45 = 0.18
 \end{aligned}$$

A	P(A)
false	0.6
true	0.4

A	B	P(B A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

B	D	P(D B)
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true	false	0.05
true	true	0.95

B	C	P(C B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

The Bad News

- Exact inference is feasible in small to medium-sized networks
- Exact inference in large networks takes a very long time
- We resort to approximate inference techniques which are much faster and give pretty good results

One last unresolved issue...

We still haven't said where we get the Bayesian network from. There are two options:

- Get an expert to design it
- Learn it from data, e.g., the same way as in the lecture on Bayes Classifier in Ch. 8.